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328672(28)

B. E. (Sixth Semester) Examination, April-May 2015

(New Scheme)

(Et & T Engg.)

OPERATING SYSTEM

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note *: Part (a) of each question is compulsory. Attempt any two parts from (b), (c) and (d) of all questions.*

1. (a) Explain the concept of file. 2

(b) Define Operating System. What are the functions provided by Operating System? 7

(c) What are the various types of operations which can

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be performed on a file? Explain with examples. 7

(d) Explain the following operating systems : (any two) 7

(i) Batch processing operating system

(ii) Multiprogramming operating system

(iii) Time sharing operating system

2. (a) Define the term Process. 2

(b) Explain life cycle of a Process. 7

(c) For the process listed below, draw a GANTT chart illustrating their execution, find the average waiting time and the average turnaround time using : 7

(i) First-come-first-served

(ii) Shortest job first

(iii) Shortest remaining time first

(iv) Round robin (quantum=2)

(v) Round robin (quantum=1)

Process	Arrival Time	CPU Burst Time
P_1	0·000	3
P_2	1·001	6
P_3	4·001	4
P_4	6·001	2

(d) Suppose we need of moving disk with 100 tracks number 0 to 99, is currently servicing a request at

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track 49 and just finished a request at track 90. If the queue of requests is kept in FIFO order— 86, 47, 91, 77, 94, 50, 02, 75, 30. Compare the performance of look and SCAN disk scheduling algorithm. 7

3. (a) What is Paging? 2

(b) What is Virtual Memory? Explain demand paging. What are the advantages of demand paging? 7

(c) What is Thrashing and how it can be controlled? Explain in brief. 7

(d) Given the reference to the following pages by a program. 7

0, 9, 0, 1, 8, 1, 8, 7, 8, 7; 1, 2, 8, 2, 7, 8, 2, 3, 8, 3

How many page faults will occur if the program has three page frames available to it and uses :

(i) FIFO replacement

(ii) LRU replacement

(iii) Optimal replacement

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4. (a) Define Deadlock. 2

(b) What is Bounded Buffer Producer-Consumer Problem? How is mutual exclusion provided in the problem? 7

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(c) What is a deadlock detection and recovery strategy? Describe deadlock detection algorithm in detail. 7

(d) Consider the following snap shot : 7

Process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P_1	1	1	1	3	1	1	4	6	1	5	3	2
P_2	1	0	1	0	1	7	5	0				
P_3	0	4	3	2	0	6	3	2				
P_4	2	0	1	4	2	6	5	6				
P_5	1	3	5	5	2	4	5	9				

(i) What is the content of need Matrix?

(ii) Is the system is in safe state?

(iii) If a new process P_1 arrives for (1, 4, 2, 3) can the request be immediately granted.

5. (a) Write short note on Device Controller. 2

(b) Explain feedback loops and its types. 7

(c) What is direct memory access? How it works? 7

(d) Write short notes on following operating systems : 7

(i) MS-DOS

(ii) Unix

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